

# Kristin Henderson

Leesburg, VA • (540) 454-8717 • kdhenderson@gmail.com

<https://kdhenderson.github.io> • <https://www.linkedin.com/in/kdhende>

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## TECHNICAL SKILLS

- **Programming:** Python, R, SAS, MATLAB, SQL, Bash
  - **Data & Analysis Libraries:** NumPy, pandas, dplyr, tidyverse, scikit-learn, XGBoost, PyTorch
  - **Machine Learning & AI:** Hugging Face, embeddings, RAG, LoRA fine-tuning, transfer learning
  - **Data Visualization:** matplotlib, seaborn, ggplot2, D3.js, Tableau, R Shiny
  - **Data Formats & Infrastructure:** Parquet, PostgreSQL, MySQL, AWS (RDS, EC2), LSF clusters
  - **Documentation & Version Control:** Git, GitHub, Jupyter notebooks, Quarto, Markdown
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## SELECTED PROJECTS

- Built a decision analysis framework in R to rank emerging brands for a national retail brokerage firm, processing a large proprietary Parquet dataset with dplyr and benchmarking rankings with ordinal logistic regression.
  - Applied time-series forecasting methods (ARMA, VAR, MLP, ensemble) in R to model multi-horizon river flow rates for resource planning.
  - Developed a Retrieval-Augmented Generation (RAG) pipeline in Python with contrastive-trained embedding models, a LoRA fine-tuned language model, and cross-encoder reranking.
  - Authored an open-source Quarto-based statistics reference linking theory to applied data science: [kdhenderson.github.io/stats-notes-for-ds](https://kdhenderson.github.io/stats-notes-for-ds)
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## WORK EXPERIENCE

### Howard Hughes Medical Institute, Janelia Research Campus, Ashburn, VA

Research Technician III, Jayaraman Lab. Dec 2021 – Present

- Built semi-automated Python pipelines to collect and process 100+ GB/day of behavioral video data for downstream statistical analysis and modeling.
- Applied statistical analysis and machine learning methods to identify patterns and behavioral structure from video tracking data.
- Collaborated with scientists and engineers to define analytical questions, assess data availability, and develop quantitative analyses and visualizations.
- Documented hardware, software, and analysis workflows to support data quality, reproducibility, and cross-team use.

Research Technician II, Keleman Lab. Jan 2017 – Dec 2021

- Led data analysis and experiments studying learning and memory; prepared datasets, figures, and results for publication.

Research Technician I, Project Technical Resources. June 2015 – Dec 2016

- Supported high-precision microscopy data collection for downstream quantitative analyses.
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## PUBLICATIONS

Meissner, G., *et al.* [including **Henderson, K**]. A split-GAL4 driver line for *Drosophila* CNS cell types. *eLife* **13**, e98405 (2025). <https://doi.org/10.7554/eLife.98405>

Lei, Z., **Henderson, K.** & Keleman, K. A neural circuit linking learning and sleep in *Drosophila* long-term memory. *Nat Commun* **13**, 609 (2022). <https://doi.org/10.1038/s41467-022-28256-1>

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## EDUCATION

Master of Science in Data Science (Expected May 2026), Southern Methodist University, Dallas, TX  
Bachelor of Science, emphasis in Biology, The Evergreen State College, Olympia, WA